or communicating nodes. The established link is shared and not dedicated.

- A 16-bit destination port, which refers to the port of the receiving device of an application process. TCP ports are defined in standard. Some of these ports are nine for transmission discard, 15 for status of network, 20 for file transfer protocol (FTP) data, 25 for SMTP e-mail applications, 80 for hyper text transfer protocol (HTTP) and 2049 for network file system (NFS).
- A 32-bit sequence number (0 to 4,294,967,295) field, which is used to assign a sequence number to each TCP pack. It is assigned to the first byte of the TCP message. The transmitter assigns the sequence number. The receiver, on reading the sequence number of the packs, ensures that all packs are received. Using a sequence number, packs received out of order are also placed in order. Using the sequence number, the receiver identifies duplicates of the packs, if any, and takes corrective measures, accordingly.
- IP sends datagrams that may be independently routed through deferent links. These, in turn, may deliver datagrams out of sequence at the receiver. Datagrams are put in order using sequence numbers assigned to these during transmission. This is done at the TCP layer. Sequence and acknowledgement numbers together ensure reliable transport of TCP, which is a reliable transport protocol, whereas IP is not.
- A 32-bit acknowledgement field, which is used to send acknowledgement of the pack received correctly. The receiver performs this function. It checks the sequence number of a received pack. If the pack is received without error, the receiver sends an acknowledgement number using the sequence number of the pack to inform the sender that the pack has been received correctly.

<table>
<tr><td colspan="8">Source Port (16 bits)</td><td>Destination Port (16 bits)</td></tr>
<tr><td colspan="9">Sequence Number (32 bits)</td></tr>
<tr><td colspan="9">Acknowledgement Number (32 bits)</td></tr>
<tr><td>Data Offset
(4 bits)</td><td>Reserved
(6 bits)</td><td>U
R
G</td><td>A
C
K</td><td>P
S
H</td><td>R
S
T</td><td>S
Y
N</td><td>F
I
N</td><td>Window (16 bits)</td></tr>
<tr><td colspan="8">Checksum (16 bits) (TCP Header+Data)</td><td>Urgent Pointer (16 bits)</td></tr>
<tr><td colspan="8">Options (Variable)</td><td>Padding (Variable)</td></tr>
<tr><td colspan="9">Data (Variable)</td></tr>
</table>

Figure 1: TCP headers

- A 4-bit offset field, which is used to indicate the number of header fields in units of 32-bit words. Offset field is also known as header length, or HLEN. If there are no padding and option fields in the header, offset field will be 0101. This means, there are five 32-bit words, or 5 x 4 = 20 bytes header. Using this information, the receiver determines the start of the data field so that data can be derived from the received pack.

Flags, window and urgent pointers are used to connect and manage the TCP connection. SYN (synchronisation), ACK (acknowledgement) and FIN (finish) flags are one-bit each. These are used to establish a TCP connection. TCP is a reliable and connection-oriented protocol. A TCP session needs a virtual connection, which is established by a process called TCP handshake. To establish a connection, the sender sends a TCP pack with SYN flag set to 1 and ACK flag set to 0. The calling program sends a TCP message through a port that has been allocated to it, like 21 for FTP connection. Initial connection requesting the TCP pack chooses a random number as the sequence number.

The connection-requesting message is a one-byte message. Thus, the random number provides the number to the calling sequence. This sequence number is called ISS (initial send sequence) for the receiver. If the random number is 250, this value is assigned to the calling sequence. If, and when, the receiver receives this pack correctly, it responds by sending a pack to the sender with both SYN and ACK flags set as 1 (SYN=1 and ACK=1).

The receiver chooses a sequence number by its random generator and sends a response message with this sequence number. If the chosen number is 350 (IRS, or initial receive sequence, number for the sender), the reply message will contain 350 in the sequence number field. The acknowledge number field contains the value 251 (meaning, the receiver is awaiting a message with sequence number 251 as it has already received message with sequence number 250).

Once the sender receives the correct pack from the receiver, the former can send another pack with ACK flag set to 1 (that is, ACK=1), with sequence number field set to 251 and acknowledge number field set to 351. The connection is then established.

After the connection is established, communicating parties can perform transmission in full-duplex mode. Connection can be terminated by any station, either sender or receiver, by sending a pack with FIN flag set (that is, FIN=1). For example, after transmitting all data, the sender can send a pack with FIN flag set to 1. When this pack is acknowledged by the receiver, connection is terminated.

During connection request phase, the initial sequence number should be so chosen that previous connections (sockets) are not confused with new connection requests (sockets). This typically happens when a host application crashes and, before the other side times out, re-establishes the crashed connection quickly by recognising that the connection request is for the old socket.